

CLOUD ENGINEERING BOOTCAMP · DAY 3

# Networking Fundamentals

## for Cloud Engineers

*From how google.com loads to how AWS VPCs are built — the networking model every cloud engineer relies on.*

### DURATION

3 hours

### PREREQUISITES

Day 1: Cloud Basics · Day 2: Linux & SSH



# Learning Objectives

*By the end of today, you will be able to:*



Explain how computer networks work



Understand the OSI & TCP/IP models



Explain IPv4 addressing & subnetting



Identify Public vs Private IP addresses



Understand CIDR notation



Explain ports and protocols



Understand DNS and DHCP



Explain routers, switches, NAT & firewalls



Understand VPNs and encrypted tunnels



Troubleshoot basic network issues



Map today's concepts directly to AWS VPC

# Today's Agenda

1

Networks, Devices & Models (OSI / TCP-IP)

2

IP Addressing, Subnetting & CIDR

3

Routing, Switching, Ports & Protocols

4

DNS, DHCP, NAT, Firewalls & VPNs

5

Troubleshooting & Hands-On Lab

6

AWS VPC Mapping & Wrap-Up

# Let's Warm Up



***“What actually happens when your computer opens google.com?”***

No wrong answers — we'll build the real, complete picture together today.

# What Is a Computer Network?

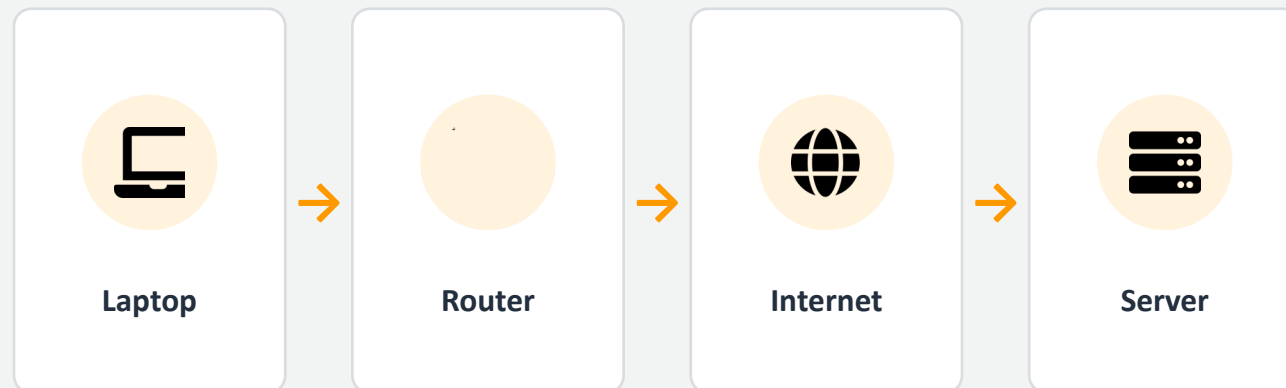
**OBJECTIVE:** Understand a network as devices connected to exchange information.

A network is two or more devices connected together so they can exchange information.

**REAL-WORLD ANALOGY:** Like a road system connecting houses to a city, then to other cities.



**COMMON MISTAKE:** Thinking WiFi and “the Internet” are the same thing — WiFi is just one way to join a network.



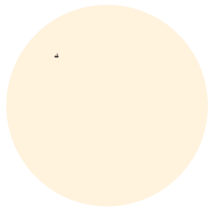
# Types of Networks

**OBJECTIVE:** Compare networks by the geographic area they cover.

| TYPE                                                                                |          | FULL NAME                 | REAL-WORLD EXAMPLE                |
|-------------------------------------------------------------------------------------|----------|---------------------------|-----------------------------------|
|    | PAN      | Personal Area Network     | Bluetooth earbuds to phone        |
|    | LAN      | Local Area Network        | Devices in one home or office     |
|    | MAN      | Metropolitan Area Network | A city's public WiFi network      |
|   | WAN      | Wide Area Network         | A company's offices across states |
|  | Internet | The network of networks   | Global — connects every WAN/LAN   |

# Network Devices

**OBJECTIVE:** Recognize the hardware that makes networks physically work.



## Router

Directs traffic between networks



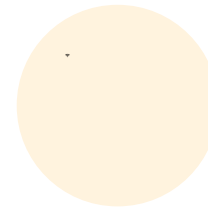
## Switch

Connects devices within one LAN



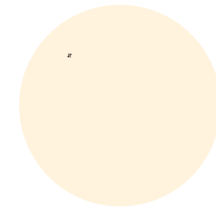
## Firewall

Filters traffic for security



## Wireless AP

Broadcasts WiFi to devices



## Modem

Connects your LAN to your ISP

**REAL-WORLD ANALOGY:** A modem is your building's mailroom entrance; a router is the front desk directing mail; a switch is the internal hallway connecting offices.

# The OSI Model — 7 Layers

**OBJECTIVE:** See the seven conceptual layers every network transmission passes through.

|   |                     |                                         |
|---|---------------------|-----------------------------------------|
| 7 | <b>Application</b>  | User-facing apps — browsers, email      |
| 6 | <b>Presentation</b> | Formatting, encryption, compression     |
| 5 | <b>Session</b>      | Opens/manages/closes connections        |
| 4 | <b>Transport</b>    | Reliable delivery — TCP / UDP           |
| 3 | <b>Network</b>      | Addressing & routing — IP               |
| 2 | <b>Data Link</b>    | Node-to-node delivery — MAC, switches   |
| 1 | <b>Physical</b>     | Cables, radio waves, electrical signals |



# OSI Layers — Analogies & Protocols

| LAYER           | REAL-WORLD ANALOGY                                 | EXAMPLE PROTOCOLS        |
|-----------------|----------------------------------------------------|--------------------------|
| 7 — Application | <i>Ordering from a restaurant menu</i>             | HTTP, HTTPS, FTP, DNS    |
| 4 — Transport   | <i>The waiter ensuring your full order arrives</i> | TCP, UDP                 |
| 3 — Network     | <i>The delivery address on a package</i>           | IP, ICMP                 |
| 2 — Data Link   | <i>The delivery truck's local route</i>            | Ethernet, MAC, ARP       |
| 1 — Physical    | <i>The literal roads the truck drives on</i>       | Cables, WiFi radio waves |



**INSTRUCTOR TIP:** Beginners typically only need Layers 3, 4, and 7 memorized well — the rest build intuition over time.

# The TCP/IP Model

**OBJECTIVE:** See the simplified 4-layer model used in real-world networking.

## Application

HTTP, DNS, SSH — combines OSI 5-7

## Transport

TCP, UDP — same as OSI Layer 4

## Internet

IP — same as OSI Layer 3

## Network Access

Ethernet, WiFi — combines OSI 1-2

## VS. OSI

TCP/IP is the practical model the real Internet runs on; OSI is the teaching reference model. Same ideas, fewer layers.

**REAL-WORLD ANALOGY:** *OSI is the detailed instruction manual; TCP/IP is the quick-start guide engineers use daily.*

# Encapsulation — Wrapping Your Data

**OBJECTIVE:** See how data gets wrapped in layers as it travels down the stack.



**REAL-WORLD ANALOGY:** Like mailing a letter inside an envelope, inside a padded package, inside a shipping box — each layer adds its own label.

# IPv4 Addressing

**OBJECTIVE:** Understand the structure of a 32-bit IPv4 address.

**192.168.1.10**

Network portion   Host portion

A 32-bit number written as four decimal numbers (0-255), separated by dots — split into a network part and a host part.

## EXAMPLES

**192.168.1.10**

A typical home network device

**10.0.0.25**

A private AWS VPC instance

**172.16.5.1**

A corporate private network device

# Public vs. Private IP Addresses

**OBJECTIVE:** Distinguish addresses reserved for private networks (RFC 1918) from public, Internet-routable ones.

## PRIVATE (RFC 1918)

Not routable on the public Internet — reused inside millions of private networks.

**10.0.0.0/8**

16.7 million addresses

**172.16.0.0/12**

1 million addresses

**192.168.0.0/16**

65,536 addresses

## PUBLIC

Globally unique and routable — how servers are reached directly from anywhere on the Internet.

**Google**

Public DNS: 8.8.8.8

**AWS**

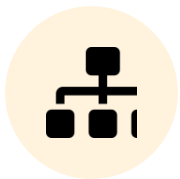
EC2 public IPs, Elastic IPs

**Cloudflare**

Public DNS: 1.1.1.1

# Subnetting Basics

**OBJECTIVE:** Break a large network into smaller, organized sub-networks.



## Network

Identifies the subnet itself

192.168.1.0



## Hosts

Usable addresses for devices

192.168.1.1 - .254



## Broadcast

Sends to every device at once

192.168.1.255



## Gateway

The router — the “way out”

192.168.1.1

**REAL-WORLD ANALOGY:** Like a neighborhood: the network is the street name, hosts are house numbers, broadcast is an announcement to the whole street, and the gateway is the road out of the neighborhood.



**COMMON MISTAKE:** Assigning a device the network address or broadcast address itself — neither can be used as a host address.

# CIDR Notation

**OBJECTIVE:** Read the /24-style shorthand that defines a network's size.

192.168.1.0/24

The number after / is how many bits are the network portion — the rest are for hosts.

| CIDR | TOTAL IPs | USABLE HOSTS |
|------|-----------|--------------|
| /24  | 256       | 254          |
| /25  | 128       | 126          |
| /26  | 64        | 62           |
| /27  | 32        | 30           |
| /28  | 16        | 14           |

# Hands-On: Splitting a /24

**OBJECTIVE:** Practice dividing 192.168.1.0/24 into smaller, equal subnets.

## 2 Subnets

**/25**

.0/25 and .128/25

*126 hosts each*

## 4 Subnets

**/26**

.0 .64 .128 .192 /26

*62 hosts each*

## 8 Subnets

**/27**

.0 .32 .64 ... .224 /27

*30 hosts each*

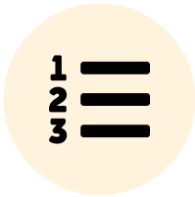


**INSTRUCTOR TIP:** Trick: each time you split in half, add 1 to the CIDR number and halve the host count.



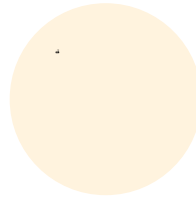
# Routing — Getting Packets There

**OBJECTIVE:** Understand how routers decide where to send each packet.



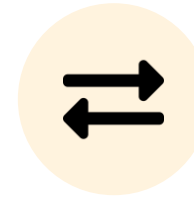
## Routing Table

A router's map of known networks & paths



## Default Gateway

Where traffic goes if no specific route matches



## Packet Forwarding

Moving a packet one hop closer to its destination

**REAL-WORLD ANALOGY:** Like a GPS: the routing table is the map, the default gateway is “when in doubt, head for the highway,” and forwarding is each turn along the way.



**COMMON MISTAKE:** Forgetting that a missing or wrong default gateway is one of the most common causes of “no internet access” errors.

# Switches vs. Routers

**OBJECTIVE:** Compare the two devices most often confused by beginners.

| CATEGORY  | SWITCH                          | ROUTER                               |
|-----------|---------------------------------|--------------------------------------|
| OSI Layer | Layer 2 (Data Link)             | Layer 3 (Network)                    |
| Purpose   | Connects devices within one LAN | Connects different networks together |
| Uses      | MAC addresses                   | IP addresses                         |
| Example   | Connecting office desks/PCs     | Connecting your LAN to the Internet  |

**REAL-WORLD ANALOGY:** A switch is the hallway connecting offices on the same floor; a router is the elevator connecting different floors (networks) entirely.

# What Is a Port?

**OBJECTIVE:** Understand how one IP address can run many different services.

A port is a numbered “door” on a device — it tells traffic exactly which service or application to reach.

**IP address** = the building

**Port** = the specific door / suite number



**COMMON MISTAKE:** Forgetting a firewall can block a port even if the IP address itself is reachable.



# Common Ports — Part 1

|                |                                         |            |                                   |                |                                                    |
|----------------|-----------------------------------------|------------|-----------------------------------|----------------|----------------------------------------------------|
| <b>20 / 21</b> | <b>FTP</b><br>File transfer             | <b>22</b>  | <b>SSH</b><br>Secure remote login | <b>23</b>      | <b>Telnet</b><br>Unencrypted remote login (legacy) |
| <b>25</b>      | <b>SMTP</b><br>Sending email            | <b>53</b>  | <b>DNS</b><br>Domain name lookups | <b>67 / 68</b> | <b>DHCP</b><br>Automatic IP assignment             |
| <b>80</b>      | <b>HTTP</b><br>Unencrypted web traffic  | <b>110</b> | <b>POP3</b><br>Retrieving email   | <b>123</b>     | <b>NTP</b><br>Time synchronization                 |
| <b>143</b>     | <b>IMAP</b><br>Retrieving email (syncd) | <b>389</b> | <b>LDAP</b><br>Directory services | <b>443</b>     | <b>HTTPS</b><br>Encrypted web traffic              |

# Common Ports — Part 2

**445****SMB**

Windows file sharing

**3389****RDP**

Windows remote desktop

**3306****MySQL**

MySQL database

**5432****PostgreSQL**

PostgreSQL database

**6379****Redis**

Redis cache/database

**8080****HTTP-Alt**

Alternate web port

**8443****HTTPS-Alt**

Alternate secure web port



**INSTRUCTOR TIP:** Memory trick: ports under 1024 are “well-known” system ports (like SSH 22, HTTP 80) — databases and custom apps usually live above 1024.

**REAL-WORLD ANALOGY:** Notice the pattern: HTTPS (443) mirrors HTTP (80); alternates just add “8” in front — 8080 and 8443.

# Common Protocols

**OBJECTIVE:** Recognize the “languages” devices use to communicate.

## TCP

Reliable, ordered delivery

## UDP

Fast, no delivery guarantee

## ICMP

Network diagnostics (ping)

## ARP

Finds MAC address for an IP

## HTTP

Unencrypted web traffic

## HTTPS

Encrypted web traffic

## FTP

File transfer

## SSH

Secure remote access

## SMTP

Sending email

## DNS

Domain name lookups

# TCP vs. UDP

**OBJECTIVE:** Compare the two core transport protocols.

## TCP

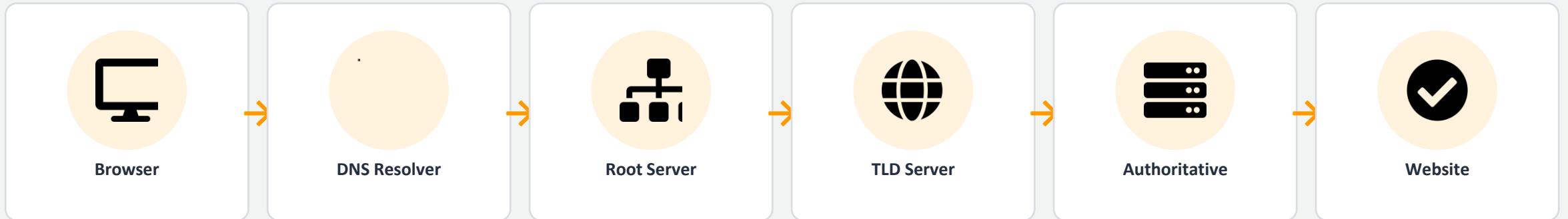
- Connection-based (handshake first)
- Guarantees delivery & order
- Used for: web, email, file transfer
- Slower due to reliability checks

## UDP

- Connectionless (no handshake)
- No delivery guarantee — “fire and forget”
- Used for: video calls, gaming, DNS
- Faster due to lower overhead

# DNS — The Internet's Phonebook

**OBJECTIVE:** Trace the full path of a domain name lookup.

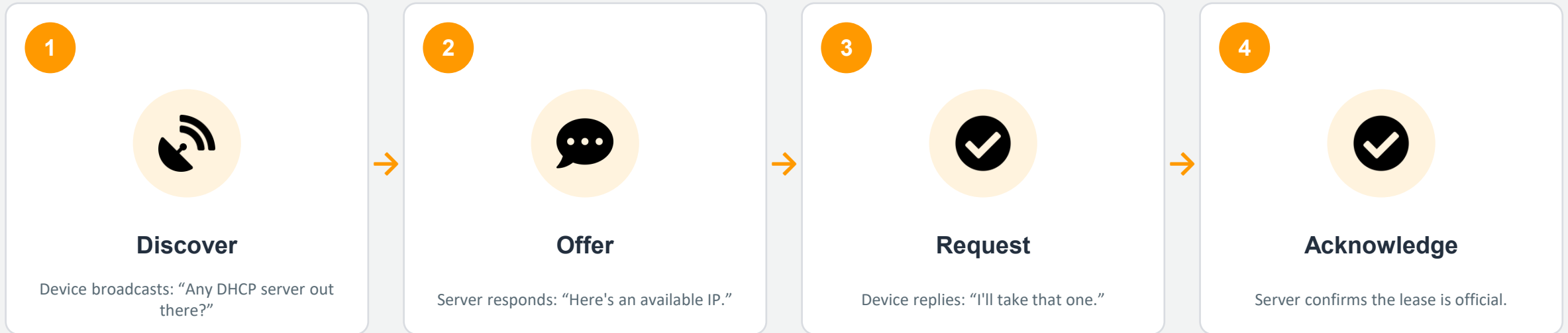


1. Browser asks the DNS Resolver (often your ISP) to look up the domain.
2. Resolver asks a Root Server: “Who manages .com?”
3. Root points to the correct TLD Server for .com domains.
4. TLD points to the domain's Authoritative Server, which holds the real IP.
5. Browser now connects directly to the website using that IP.



# DHCP — Automatic IP Assignment

**OBJECTIVE:** See how a device automatically receives an IP address when it joins a network.



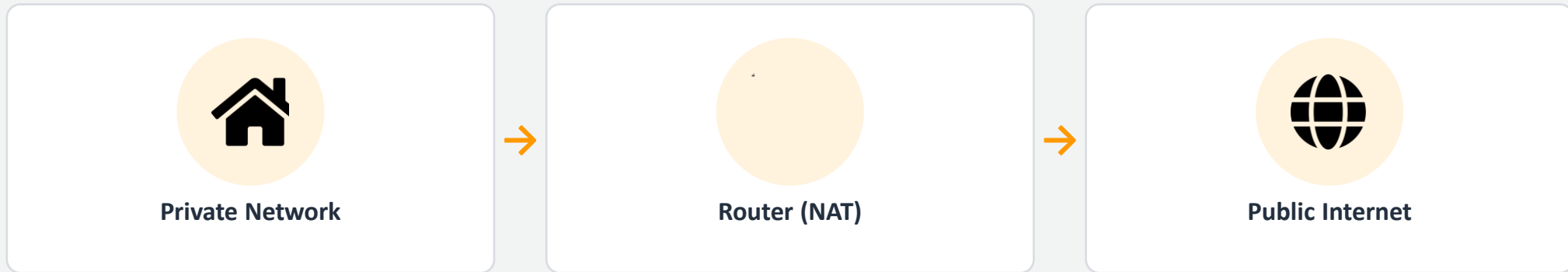
*Also known as "DORA" — Discover, Offer, Request, Acknowledge.*



**INSTRUCTOR TIP:** Recommend the mnemonic "Don't Offer Rotten Apples" (Discover, Offer, Request, Acknowledge) to help it stick.

# NAT — Network Address Translation

**OBJECTIVE:** See how many private devices share one public IP address.



## PAT

Port Address Translation — many devices share one public IP, tracked by port

## SNAT

Source NAT — rewrites the sender's private IP to a public one

## DNAT

Destination NAT — forwards public traffic to a private internal server

# Firewalls — Filtering Traffic

**OBJECTIVE:** Understand how firewalls decide what traffic to allow or block.

## Stateful

Tracks the full conversation — automatically allows return traffic for a connection it already approved.

## Stateless

Evaluates every single packet independently, with no memory of prior traffic — rules must cover both directions.

## REAL-WORLD EXAMPLES

### Windows Firewall

Stateful, host-based

### AWS Security Groups

Stateful, attached to EC2 instances

### AWS NACLs

Stateless, attached to subnets

# VPN — Encrypted Tunnels

**OBJECTIVE:** See how a VPN protects traffic as it crosses an untrusted network.



**REAL-WORLD ANALOGY:** Like an armored, opaque tunnel through a public highway — outside observers see traffic moving, but can't see or touch what's inside.



**INSTRUCTOR TIP:** AWS Site-to-Site VPN connects an on-premises network securely to a VPC over the public Internet.

# Essential Troubleshooting Commands

**OBJECTIVE:** Know which command to reach for when something isn't working.

## **ping**

Tests basic connectivity to a host

## **tracert / traceroute**

Shows the path packets take, hop by hop

## **ipconfig / ifconfig / ip addr**

Shows your device's own IP configuration

## **netstat / ss**

Lists active connections & listening ports

## **arp**

Shows local IP-to-MAC address mappings

## **route**

Displays the local routing table

## **nslookup / dig**

Looks up DNS records for a domain

## **curl**

Makes a direct request to a URL

## **telnet / Test-NetConnection**

Tests if a specific port is reachable

# Practical Lab — Diagnose Your Network

**OBJECTIVE:** Run real commands to inspect and test your own network connection.

PowerShell / Terminal

```
$ ipconfig
  (view your IP, gateway & DNS server)
$ ping google.com
  (test basic connectivity)
$ tracert google.com
  (see every hop along the way)
$ nslookup google.com
  (confirm DNS resolution)
$ curl google.com
  (request the actual webpage)
```

## EXPECTED OUTPUTS

- A local IPv4 address & default gateway
- 4 successful replies with a low ms time
- A list of routers between you and Google
- Google's public IP address(es)
- Raw HTML starting with <!doctype html>



**INSTRUCTOR TIP:** Use Mac/Linux: ping, traceroute, nslookup, curl.  
Windows: tracert instead of traceroute.

# How This Maps to AWS Networking

**OBJECTIVE:** See today's concepts become real AWS VPC building blocks.



**Internet Gateway** = your router's connection to your ISP

**VPC** = your own private, isolated network (like a whole office building)

**Subnet** = a floor within that building (with its own CIDR block)

**Route Table** = the routing rules deciding where each floor's traffic goes

# AWS Hands-On Lab — Your First Network

**OBJECTIVE:** Build a simple, working network in the AWS Console — no coding required.

## 6 SIMPLE STEPS

- 1 Create a VPC**  
CIDR block 10.0.0.0/16
- 2 Create a subnet**  
10.0.1.0/24, inside the VPC
- 3 Create & attach an Internet Gateway**  
Gives the VPC a path to the Internet
- 4 Update the route table**  
Add route 0.0.0.0/0 → Internet Gateway
- 5 Launch a t2.micro EC2 instance**  
Place it in your new public subnet
- 6 Connect and test**  
SSH into the instance, then ping google.com



### You'll Know It Worked When...

Your EC2 instance gets a public IP, you can SSH into it, and it can successfully ping the public Internet.

~30–45 minutes · AWS Free Tier eligible



**INSTRUCTOR TIP:** Terminate the EC2 instance when you're done — stay inside the Free Tier and avoid surprise charges.



# Summary — What We Covered

- ✓ Networks connect devices to exchange data, at different scales (PAN to Internet)
- ✓ OSI & TCP/IP models describe how data is layered and transmitted
- ✓ IPv4 addresses, subnetting & CIDR define how devices are organized and addressed
- ✓ Ports & protocols identify which service traffic is meant for, and how it's delivered
- ✓ DNS, DHCP, NAT, firewalls & VPNs are core services enabling and securing connectivity
- ✓ Troubleshooting tools help diagnose problems layer by layer
- ✓ Every concept maps directly onto AWS VPC networking

# Avoid These Networking Pitfalls



## CIDR Confusion

Thinking a higher CIDR number (e.g. /28) means a bigger network than a lower one (e.g. /24).



## Missing Default Gateway

Assuming devices can reach the Internet without a correctly set gateway.



## Overly Open Firewalls

Opening all ports “just to make it work” instead of allowing only what's needed.



## Confusing NAT with Security

Believing NAT alone makes a private network secure, rather than just address-hidden.



**INSTRUCTOR TIP:** When troubleshooting, check in order: cable/WiFi → IP config → gateway → DNS → firewall/ports.

## Quiz — Questions 1–5

1

What does a router primarily do?

2

Which OSI layer handles physical cables and radio signals?

3

Which model is actually implemented on the real Internet: OSI or TCP/IP?

4

What does the network portion of an IP address identify?

5

Which of these is a private IP range: 8.8.8.8 or 192.168.1.1?

## Quiz — Questions 6–10

6

In CIDR notation, does a larger number after the slash mean a bigger or smaller network?

7

What port does HTTPS use by default?

8

Which transport protocol guarantees delivery: TCP or UDP?

9

What type of DNS server holds the actual, final answer for a domain?

10

What does the 'D' in DORA (DHCP) stand for?

## Quiz — Questions 11–15

11 What does NAT allow multiple private devices to share?

12 What's the difference between a stateful and a stateless firewall?

13 What does a VPN primarily protect while data crosses an untrusted network?

14 Which command shows the path (hops) packets take to a destination?

15 In AWS, which component plays the same role as a home router's link to your ISP?

# Quiz — Answer Key

1. Directs traffic between different networks

2. Layer 1 — Physical

3. TCP/IP

4. Which network a device belongs to

5. 192.168.1.1 (private)

6. Smaller network

7. Port 443

8. TCP

9. Authoritative server

10. Discover

11. One public IP address

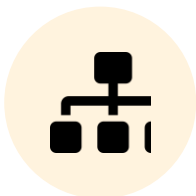
12. Stateful remembers connections; stateless checks every packet

13. The content/data itself

14. traceroute / tracert

15. Internet Gateway

# Homework



## Subnet Practice

Split 10.0.0.0/16 into 4 equal subnets and write out each CIDR block.



## Command Practice

Run all 5 lab commands against 3 different websites and record the results.



## Read: AWS VPC Docs

Skim the AWS VPC “Getting Started” guide to preview tomorrow's hands-on lab.

# Reference & Cheat Sheets

*Quick-reference material to keep handy as you continue building AWS skills.*





# Networking Cheat Sheet

## IP Address

Unique identifier for a device on a network

## Subnet

A smaller, segmented piece of a larger network

## CIDR

Shorthand for network size, e.g. /24

## Gateway

The router that connects a subnet to other networks

## DNS

Translates domain names into IP addresses

## DHCP

Automatically assigns IP addresses to devices

## NAT

Lets many private devices share one public IP

## Firewall

Filters traffic based on rules

## VPN

Encrypts traffic across an untrusted network

## Port

Identifies which service traffic is meant for

## Router

Connects different networks together

## Switch

Connects devices within the same network

# Port Cheat Sheet

|              |               |              |                    |               |                         |
|--------------|---------------|--------------|--------------------|---------------|-------------------------|
| 20/21<br>FTP | 22<br>SSH     | 23<br>Telnet | 25<br>SMTP         | 53<br>DNS     | 67/68<br>DHCP           |
| 80<br>HTTP   | 110<br>POP3   | 123<br>NTP   | 143<br>IMAP        | 389<br>LDAP   | 443<br>HTTPS            |
| 445<br>SMB   | 3306<br>MySQL | 3389<br>RDP  | 5432<br>PostgreSQL | 6379<br>Redis | 8080/8443<br>HTTP/S Alt |



**INSTRUCTOR TIP:** 22=SSH, 80/443=Web, 53=DNS — memorize these three first; the rest follow with repetition.

# CIDR Cheat Sheet

| CIDR | TOTAL IPs | USABLE HOSTS |
|------|-----------|--------------|
| /16  | 65,536    | 65,534       |
| /20  | 4,096     | 4,094        |
| /24  | 256       | 254          |
| /25  | 128       | 126          |
| /26  | 64        | 62           |
| /27  | 32        | 30           |
| /28  | 16        | 14           |
| /29  | 8         | 6            |
| /30  | 4         | 2            |

# OSI Cheat Sheet

|   |              |                       |
|---|--------------|-----------------------|
| 7 | Application  | HTTP, DNS, FTP        |
| 6 | Presentation | SSL/TLS, encoding     |
| 5 | Session      | Session establishment |
| 4 | Transport    | TCP, UDP              |
| 3 | Network      | IP, ICMP              |
| 2 | Data Link    | Ethernet, MAC, ARP    |
| 1 | Physical     | Cables, radio waves   |

*Mnemonic: Please Do Not Throw Sausage Pizza Away (bottom → top)*

# TCP vs. UDP — Quick Reference

| CATEGORY    | TCP                          | UDP                      |
|-------------|------------------------------|--------------------------|
| Connection  | Connection-based (handshake) | Connectionless           |
| Reliability | Guaranteed delivery & order  | No guarantee             |
| Speed       | Slower (more overhead)       | Faster (less overhead)   |
| Common Uses | Web, email, file transfer    | Video calls, gaming, DNS |

# Useful Commands Reference

**ping <host>**

Test connectivity

**tracert/tracert <host>**

Trace the path taken

**ipconfig / ifconfig / ip addr**

Show local IP config

**netstat -an / ss -tulpn**

List active connections

**arp -a**

Show local ARP table

**route print / ip route**

Show routing table

**nslookup <domain>**

Basic DNS lookup

**dig <domain>**

Detailed DNS lookup

**curl <url>**

Request a URL directly

**telnet <host> <port>**

Test if a port is open

**Test-NetConnection <host>**

Windows port/connectivity test



# Great Work Today!

*You now understand how networks, IP addressing, and the Internet itself actually work — the foundation for everything in AWS VPC.*